ELITE IGCSE ACADEMY

Student Past Paper Solutions

May-Nov 2020 P1H Worked Solutions

MayNov 2020 | P1H

31 questions ▪ question image followed by worked solution

Prepared by

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Student Edition

Worked answers included

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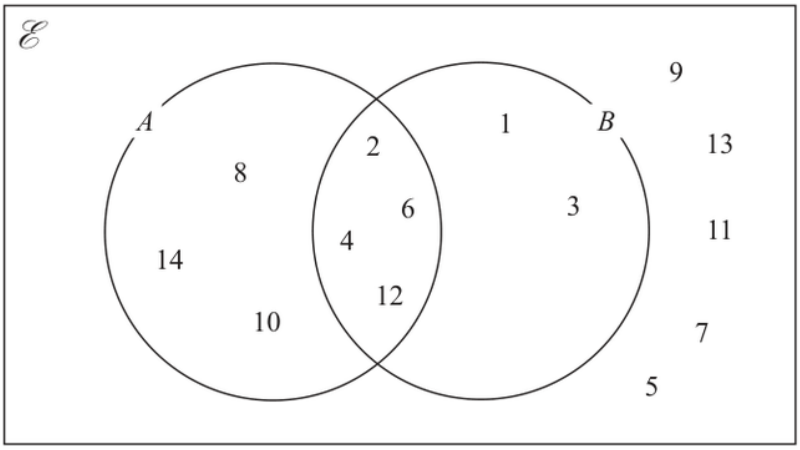
PAST PAPER QUESTION Q#: 1 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

1 The numbers from 1 to 14 are shown in the Venn diagram.



(a) List the members of the set $A \cap B$

....

(1)

(b) List the members of the set B'

....

(1)

A number is picked at random from the numbers in the Venn diagram.

(c) Find the probability that this number is in set A but is **not** in set B.

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

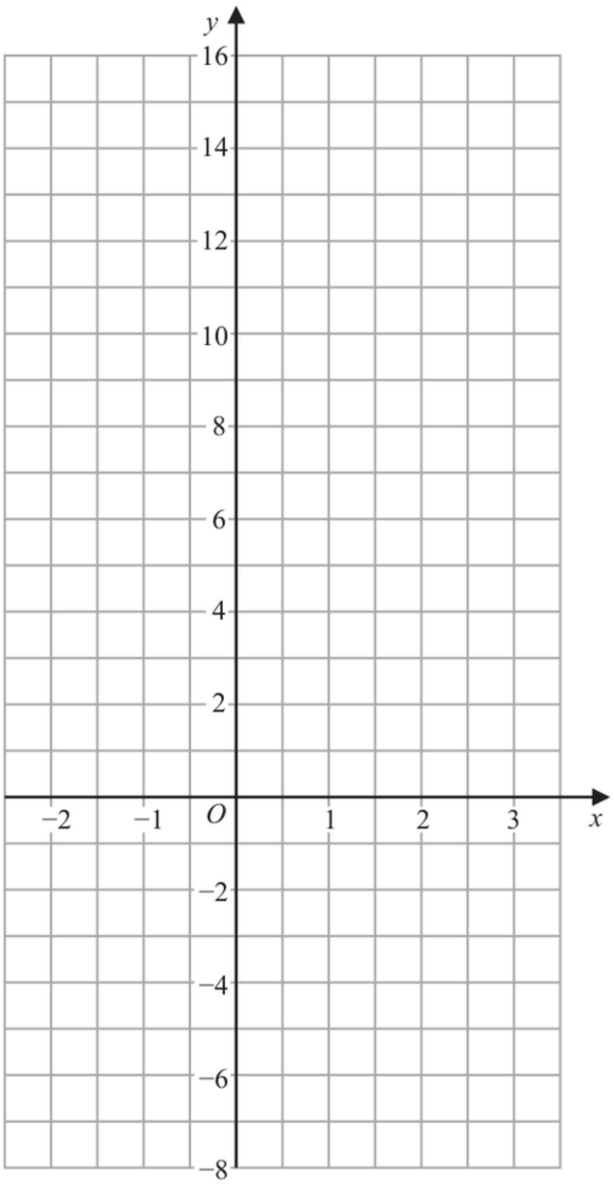
PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Linear Graphs (y = mx + c)**

May-Nov 2020 P1H - MayNov 2020 | P1H

3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Linear Graphs ($y = mx + c$)**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9
a mean of 11

Find the value of x and the value of y.

x = ...

y = ...

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

5 (a) Write 5.7×10^{-3} as an ordinary number.

....
(1)

(b) Write 800 000 in standard form.

....
(1)

(c) Work out $\dfrac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

....
(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.
Give your answer in seconds, correct to the nearest second.

.... seconds

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 7 Marks: 8

TOPIC: **Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$x =$ ....
(3)

(b) Factorise fully $16m^3g^3 + 24m^2g^5$

....
(2)

(c) (i) Factorise $y^2 - 2y - 48$

....
(2)

(ii) Hence, solve $y^2 - 2y - 48 = 0$

....
(1)

(Total for Question 7 is 8 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 8

TOPIC: **Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

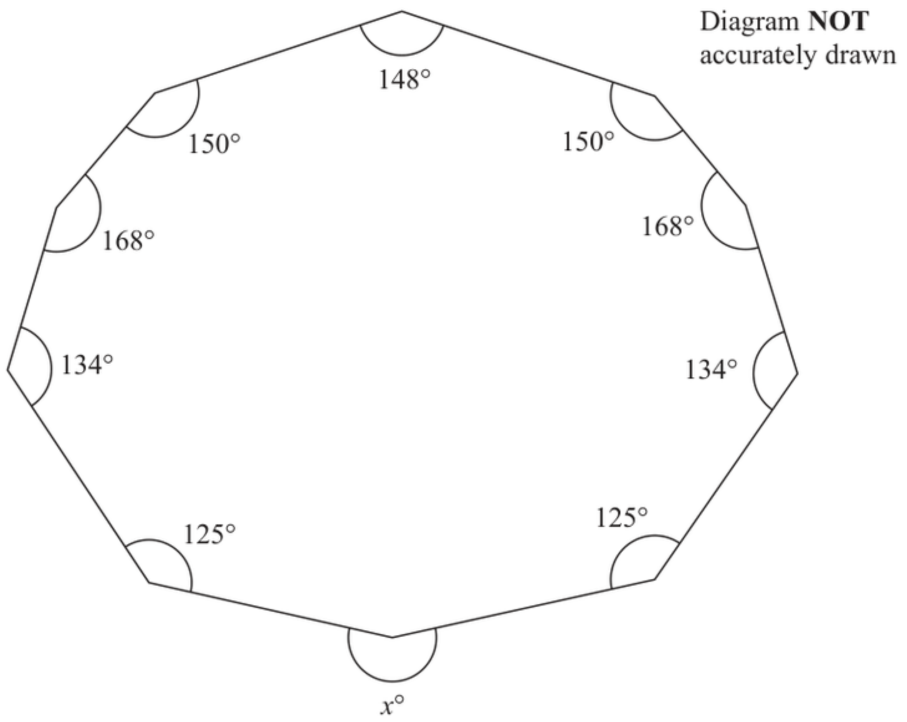
PAST PAPER QUESTION

Q#: 8 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

8 Here is a 10-sided polygon.



Work out the value of x.

$x =$ ....

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 9 Marks: 3

TOPIC: **Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

.... rupees

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

10 $A = 2 \times 3^{43}$
$B = 16 \times 3^{37}$

(a) Find the highest common factor (HCF) of A and B.

....
(1)

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

....
(2)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 11 Marks: 6

TOPIC: **Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

11 The diagram shows trapezium *ABCD* in which *BC* and *AD* are parallel.

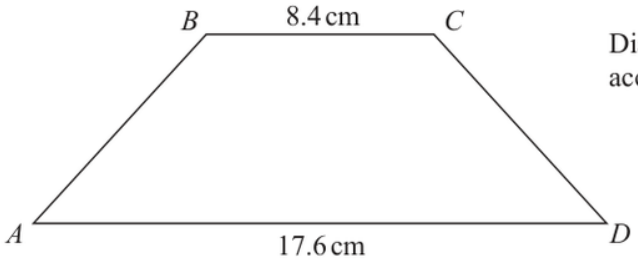

Diagram **NOT** accurately drawn

The trapezium has exactly one line of symmetry.

$BC = 8.4$ cm
$AD = 17.6$ cm

The trapezium has area $179.4\,\text{cm}^2$

Work out the size of angle *ABC*.
Give your answer correct to 1 decimal place.

.... °

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 6

TOPIC: **Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

12 Solve the simultaneous equations

$$7x - 2y = 34$$
$$3x + 5y = -3$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 13 Marks: 3

Topic: **Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

13 Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of x% per year.

At the end of 6 years, there is a total of \$8877.62 in the account.

Work out the value of x.
Give your answer correct to 2 decimal places.

$x =$ ....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

14 F is inversely proportional to the square of v.

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v.

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

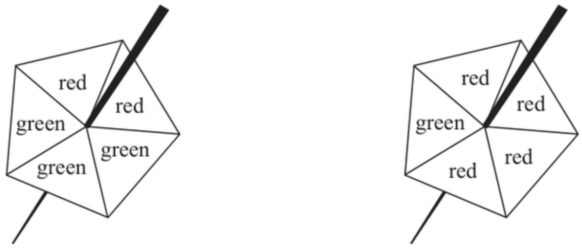
PAST PAPER QUESTION

Q#: 15 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

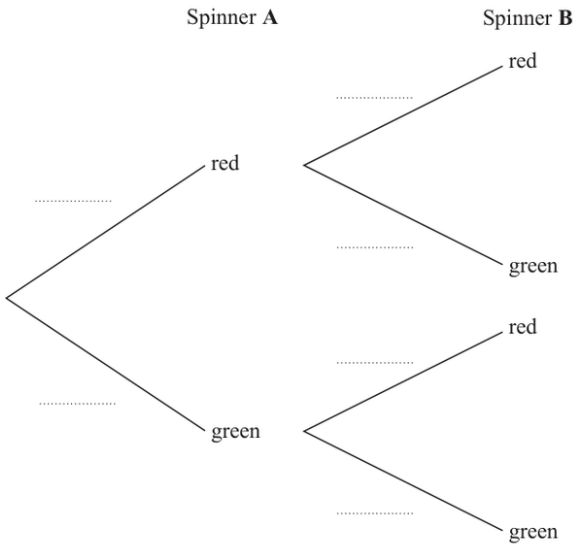
15 Harry has two fair 5-sided spinners.



Spinner **A** Spinner **B**

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that at least one of the spinners will land on green.

....

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

16

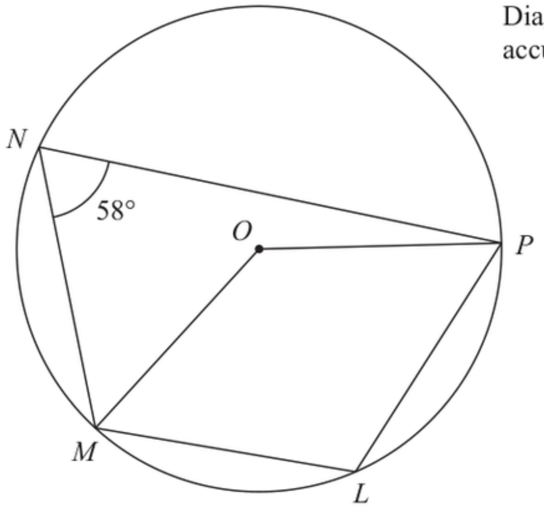

Diagram **NOT** accurately drawn

L, M, N and P are points on a circle, centre O

Angle $MNP = 58°$

(a) (i) Find the size of angle MLP

.... °

(ii) Give a reason for your answer.

....

(2)

(b) Find the size of the reflex angle MOP

.... °

(2)

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 17 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3})\,m^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

.... kg/m^3

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 18 Marks: 5

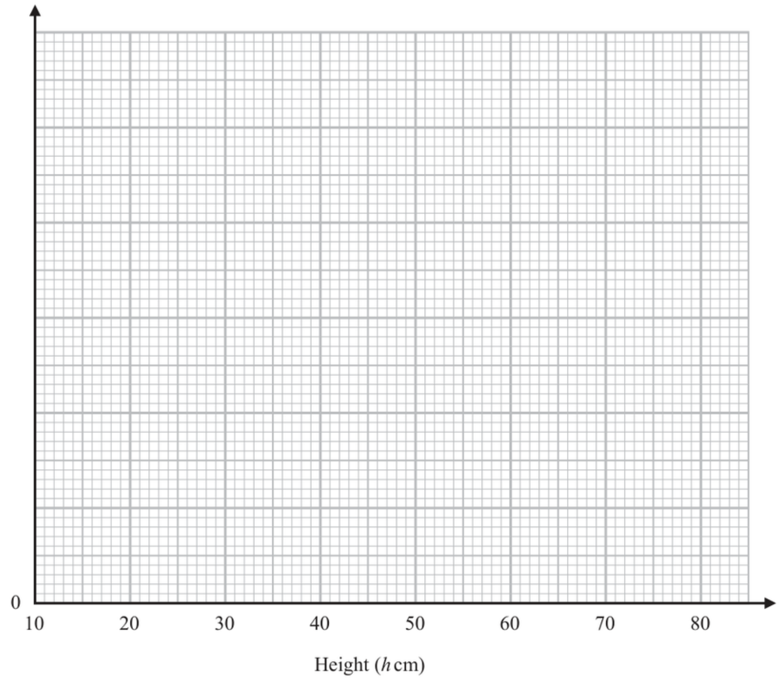
TOPIC: **Histograms**

May-Nov 2020 P1H - MayNov 2020 | P1H

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	**Frequency**
$10 < h \leqslant 20$	35
$20 < h \leqslant 35$	45
$35 < h \leqslant 50$	75
$50 < h \leqslant 70$	40
$70 < h \leqslant 80$	8

(a) On the grid, draw a histogram for this information.



(3)

(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

....

(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

19 Without using a calculator, rationalise the denominator of $\dfrac{6}{3-\sqrt{7}}$

Simplify your answer.
You must show each stage of your working.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 20 **Marks: 3**

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3
Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

.... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3
Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

.... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

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as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

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The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.............................. , ...)

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 22 Marks: 4

Topic: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

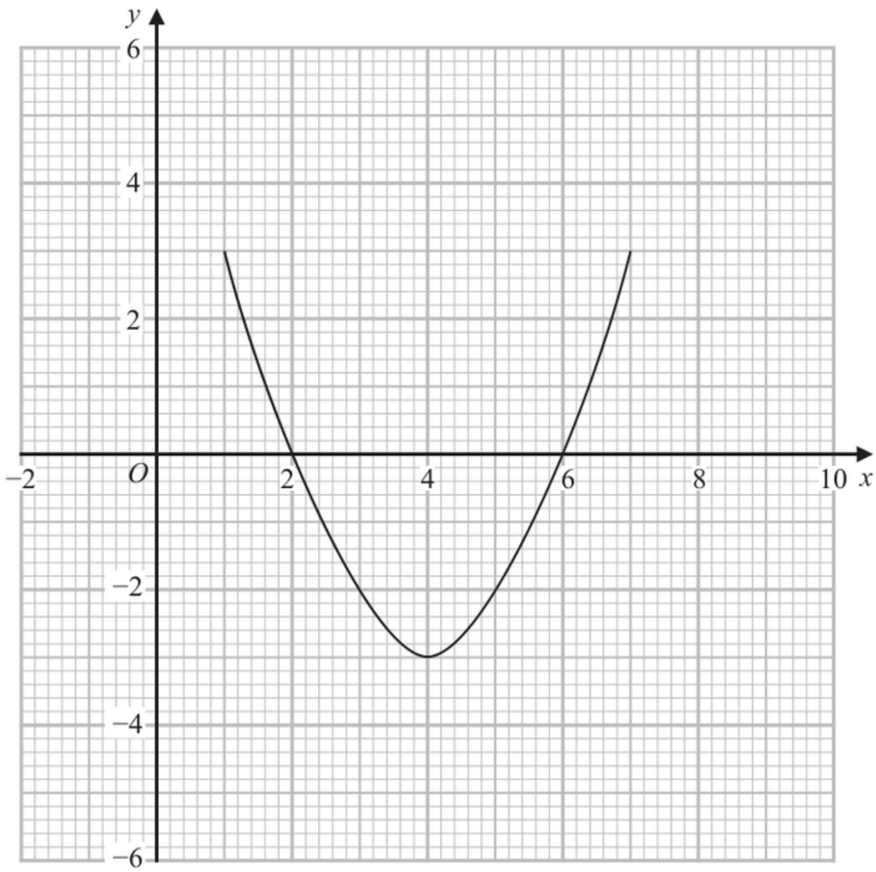
25 The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

....

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 25 Marks: 3

Topic: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

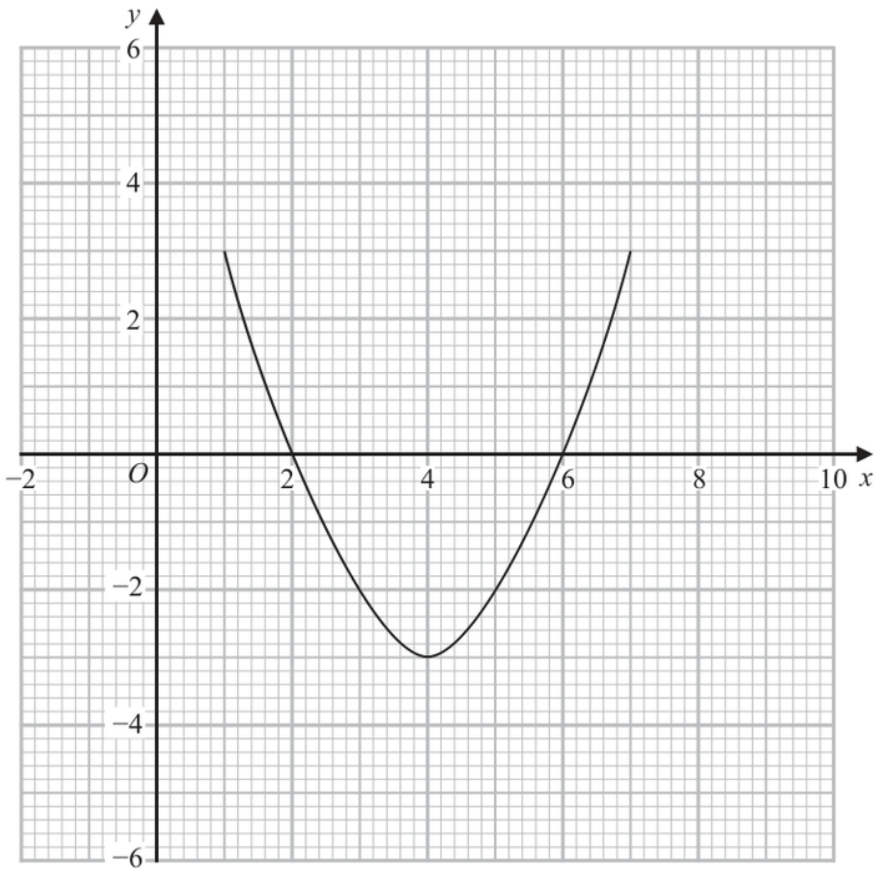
25 The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

....

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.